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Software Design Document

Of

EEE Components Storage and Management System



**U R Rao Satellite Centre (URSC) Old Airport Road Vimanapura Post,
Bengaluru , Pin code - 560017**



Neemus Software Solutions Private Limited

Plot No.2, Jagruthi Shines, RR Nagar, Old Bowenpally, Hyderabad - 500011

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Sl no.	Name	Designation	Division/Lab/Firm	Remarks
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2.	Abhishek Singh Yadav	Scientist/Engineer 'SF'	ISRO	
3.	Avneesh Gupta	CEO	Neemus	
4.	Bharath	Software Engineer	Neemus	
5.	Akhilesh	Software Engineer	Neemus	
6.	Bhavya Sri	Software Engineer & QA	Neemus	
7.	Goutham Krishna	Software Engineer & QA	Neemus	
8.	Dhanush	Software Engineer & QA	Neemus	

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1. Introduction

The **EEE Components Management System (EEE_CMS)** is implemented in **ERPNext**, designed to manage Electrical, Electronic & Electromechanical (EEE) components for spacecraft projects at URSC.

The system provides a **web-based, modular, and user-friendly interface** for:

- Inventory tracking of EEE components.
- Project card creation and management.
- Component issuance and FIFO traceability.
- Integration with external systems like **FMS** and **PCBMS**.

This SDD defines the **design, architecture, and implementation strategy** of the ERPNext-based system, including modules, workflows, data models, user roles, and integration points.

2. Purpose

The purpose of this SDD is to provide a **technical blueprint** for implementing the EEE Components Management System in ERPNext. It guides developers, testers, and stakeholders on:

- System **modules and workflows** for component management.
- **Data models (DocTypes) and their relationships.**
- **User roles, permissions, and access control.**
- Integration with external systems such as FMS and PCBMS.
- **Customization strategies** in ERPNext for URSC-specific requirements.

The design ensures the system is **scalable, maintainable, secure, and aligned with ERPNext best practices.**

3. Scope Purpose

The scope of this SDD includes:

- **Implementation of ERPNext DocTypes for:**
 - EEE Components
 - Project Cards
 - Stock Management (Warehouse, Batch, Serial Numbers)
 - Component Issuance and Returns
- **Workflows for:**
 - Component requests

SDD

- Approvals and issuance
- Traceability and FIFO compliance
- **Reports and dashboards** for inventory status, project usage, and compliance tracking.
- **Integration** with FMS and PCBMS for data synchronization.
- **Role-based access control** for different user types: Inventory Manager, Project Engineer, Auditor.
- **Out of scope:**
 - Legacy system features not relevant to ERPNext.
 - Non-ERPNext software components, except for data migration.

4. Identification

Give identification details of this document.

Identification Document Number: EEE_CMS_SDD_1.0

Baseline No: 1.0

Release No: 1.1

5. Abbreviations and Acronyms

This section typically includes common abbreviations and acronyms used throughout the document. Here's a possible table format:

ALT	Alternate
SDD	Software Design Document
KPI	Key Performance Indicator
BOM	Bill of Material
CMG	Component Management Group
CPD	Component Procurement Division
CQCD	Component Quality Control Division
DPA	Destructive Project Analysis
DPD	Deputy Project Director
ECAD	Electronic CAD Facility
EEE	Electrical, Electronic & Electromechanical
ERP	Enterprise Resource Planning
ESD	Electrostatic Discharge
FA	Failure Analysis
FIFO	First In First Out

SDD

FMS	Fabrication management System
HMC	Hybrid Micro Circuits
HQCD	HMC Quality Control Division
ICSDBS	Integrated Components Services Database System
KOHM	Kilo Ohm
M/R	Main & Redundant
MDB	Master Database
MIRV	Material Inspection cum receipt Voucher
MOQ	Minimum Order Quantity
Non – SMT	Non-Surface mount
PCBMS	PCB Management System
PD	Project Director
PMO	Program Management Office
PO	Purchase Order
PPM	Parts Per Million
QA	Quality Assurance
RDC	Request for Design Change
RHEL OS	Red Hat Enterprise Linux Operating System
SAT	Satellite
SDD	S/W Design Documentation
SIV	Store Issue Voucher
SMEG	System External Manufacturing Group
SMT	Surface mount
SP	Spare
SRN	Store Return Note
SRV	Store Requirements Specification
SRS	Software requirements Specification
TDP	Technology Development Project
TO-SCR	To Screen
TS	Test Select
URSC	U R Rao Satellite Centre

6. Referenced Documents

6.1 External References

- ERPNext Documentation (<https://docs.erpnext.com/>)
- Frappe Framework Documentation (<https://frappeframework.com/>)
- Python Official Documentation (<https://docs.python.org/>)
- MariaDB / PostgreSQL Documentation

6.2 Internal References

- Statement of Work (SoW) for EEE Components Storage and Management System
- Software Design Document (SDD) for EEE Components Storage and Management System
- URSC IT Standards and Deployment Guidelines

7. System Overview

7.1 Background

The U R Rao Satellite Centre (URSC) requires a modern system to manage Electrical, Electronic, and Electromechanical (EEE) components used in spacecraft projects. The new system is being developed on **ERPNext (open-source ERP)**, leveraging the **Frappe framework**, **MariaDB/PostgreSQL**, and **Python**.

This ERPNext-based solution provides a **web-based, modular, and scalable environment** to manage the complete lifecycle of EEE components — from procurement and storage to issuance, traceability, and project card management. The platform ensures compliance with URSC IT standards, enables interoperability with related systems, and eliminates recurring licensing costs.

7.2 High-Level Goals

The key objectives of the new ERPNext-based EEE Component Management System are:

- **Modernization:** Implement an open-source ERP solution (ERPNext) with scalable, modular architecture.
- **Integration:** Enable seamless interoperability with PCBMS (PCB Management System) and FMS (Fabrication Management System).
- **Inventory Accuracy:** Implement FIFO logic, expiry tracking, reliving workflows, and redundancy management.
- **Traceability:** Maintain complete audit logs for SRV (Store Receipt Voucher), SRN (Store Return Note), SIV (Store Issue Voucher), transfers, and approvals.

SDD

- **Configurability:** Support project-wise card configuration (Project → Subsystem → Package → Card → BOM) with approval workflows (ECAD, CPD).
- **Scalability:** Handle up to **10,000 users** with **300 concurrent sessions**.
- **Performance:** Ensure report response times of ≤ 10 seconds for standard reports and ≤ 5 minutes for bulk reports.
- **Compliance:** Follow 100% open-source licensing (ERPNext, Python, MariaDB/Postgres, Ubuntu/RHEL).

7.3 Target Users

The ERPNext-based system is designed for multiple stakeholders across URSC and collaborating centres:

- **Stores & Logistics Teams:** Manage SRV (receipts), SRN (returns), SIV (issuance), and inter-program stock transfers.
- **Subsystem Engineers:** Configure project BOMs, request components, approve card configurations, track availability.
- **Quality Assurance (QA/ECAD/HQCD/CQCD Teams):** Approve BOMs, handle reliving/screening requests, validate project cards.
- **Project Managers (PD/DPD):** Oversee project configurations, approvals, and inventory usage.
- **Program Management Office (PMO):** Access dashboards, KPI reports, and stock summaries.
- **External Centre Users (LEOS, IISU, LPSC, SAC):** Request and track components for inter-centre collaboration.
- **Developers & Administrators:** Maintain ERPNext, configure workflows, update master data, and generate new reports.

7.4 Context Diagram

At a high level, the ERPNext-based **EEE Component Management System** interacts with:

- **Core ERPNext Modules (within the system):**
 - Inventory Management (SRV, SIV, SRN, Transfers, Verification).
 - Project & Card Configuration (Projects, Subsystems, Packages, Cards, BOMs).
 - Master Data (MDB part numbers, vendors, package styles, quality levels).
 - Special Modules (Radiation Data, FA – Failure Analysis, DPA – Destructive Physical Analysis)
 - Reports & Dashboards (MIS/KPIS).
- **External Systems (integrated with ERPNext):**
 - **PCBMS (PCB Management System)** – PCB design and fabrication workflows.
 - **FMS (Fabrication Management System)** – Card fabrication and production workflows.

➤ **Users (via Web Interface):**

- Engineers, QA teams, managers, and stores personnel (role-based access).
- Remote users from collaborating centres (secure access).

The system operates on URSC's **virtual machine infrastructure**, with two redundant servers running Ubuntu/RHEL, ensuring **high availability and ≤1 hour downtime** in case of failures.

8. Detailed Software Design

8.1 Module Design

The EEE Components Storage and Management System is designed as a modular ERPNext-based solution. Each module addresses a specific functional requirement outlined in the Statement of Work (SoW) and ensures complete lifecycle management of EEE components with traceability, approval workflows, and integration.

8.1.1 Inventory Module

The Inventory Module uses ERPNext's Stock Entry doctype as the base to manage all store-related transactions for EEE components at URSC. It covers the complete lifecycle: receiving, issuing, returning, and transferring stock

8.1.1.1 Component Management (Extended Item Module)

The Component Management module extends ERPNext's Item and Item Supplier doctypes to handle electronic/electrical components with controlled Basic Part Number (BPN) generation, classification by Component Group, and linkage through a common Link Part Number. It ensures supplier mapping, BOM consistency, and controlled component traceability.

1. Doctype Customization

Base Doctypes

- **Item (Extended as Component)** – base ERPNext Item customized for electronic/electrical components.
- **Item Supplier (Extended as Component Supplier)** – supplier mapping with auto-generated BPN.
- **Item Group (Extended as Component Group)** – hierarchical classification of components.
- **Package Style (Custom)** – approved package style catalog (e.g., DIP, SMD).
- **Footprint (Optional Custom)** – footprint library for PCB design integration.
- **Link Part Number (Custom)** – ties all supplier-specific BPNs together.

Steps

1. Go to **Settings → Customization → Customize Form**.
2. Select **Item** as the doctype to customize.
3. Create a **new section** (e.g., *Component Details*) to group component-specific fields.
4. Add fields for component code, component name, component group, package

in this place you
component to base
part number

here in the place of supplier part number to change
SRV part number remove from here BPN

style, footprint, link part number, and default UoM.

5. Select **Item Supplier** doctype and add fields for supplier name and supplier part number.
6. Configure the system to **Enter Basic Part Number (BPN)** as a combination of component code + supplier part number.
7. Save and reload the forms to apply changes

remove this we are not generat
BPN remove and table only two fields
1)supplier
2)SRV part number

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings → Workflow → New Workflow**.
2. Set:
 - **Workflow Name:** Component Management Workflow
 - **Document Type:** Item
 - **Condition:** doc.is_component == 1 (custom flag if needed)
3. Click **Save**.

2) Add Workflow States

- Draft
- QA Review
- Approved
- Released

3) Add Workflow Transitions

a) Draft → QA Review

- **Action:** Send for QA Review
- **Allowed Role:** Component Engineer
- **Next State:** QA Review

b) QA Review → Approved

- **Action:** Approve
- **Allowed Role:** QA Approver
- **Next State:** Approved

c) Approved → Released

- **Action:** Release
- **Allowed Role:** System Admin
- **Next State:** Released

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply to all components created in the Item doctype.

3. Permission Rules

- **Component Engineer:** Create/Edit Components, add suppliers.
- **QA Approver:** Validate package style, footprint, and link part number.
- **Procurement Officer:** Use component and supplier mappings in purchase orders.
- **System Admin:** Manage groups, package styles, and UoM defaults.

Implement these permissions via the **Role Permission Manager**.

4. Supplier Mapping & BPN Generation

- When a supplier is added under **Component Supplier**, the system should automatically generate a **Basic Part Number (BPN)** using the format: **BPN = Component Code + Supplier Part Number**.
- Each supplier's BPN is tied to the same **Link Part Number** for that component.
- This ensures all supplier variations for a component point back to a single unified reference.

5. Reports

a) Component Register

- Type: List Report
- Shows: Component Code, Name, Group, UoM, Suppliers

b) Supplier Mapping Report

- Type: Query Report
- Shows: Supplier ↔ Component ↔ BPN mapping

c) Link Part Consistency Report

- Type: Query Report
- Verifies that all supplier parts for a component are correctly mapped under the same Link Part Number

d) Component Group Report

- Type: Tree/List Report
- Shows classification hierarchy with count of components in each group

6. Integration Hooks

- **Item Master (ERPNext)** → Components extend the base Item master with additional fields.
- **Purchase Orders** → Auto-fetch supplier part numbers with mapped BPNs when creating POs.
- **BOM (Bill of Materials)** → Use **Link Part Number** as the common reference to consolidate supplier variations.
- **QA Module** → QA Approver validates package style, footprint, and classification before approval.
- **Stores / SRV Module** → SRV validates received part numbers against component mappings before accepting into stock.

8.1.1.2 SRV (Store Receipt Voucher) Module

The **SRV module** extends ERPNext's **Stock Entry (Material Receipt)** doctype to capture receipt of EEE components into URSC stores. It ensures traceability, QA compliance, and integration with project configurations.

1. Doctype Customization

Base Doctype

- **Stock Entry** (Material Receipt Purpose).
- Extend this with **Custom Fields**.

Steps

Add all fields of SRV (Enter,post,bin)

1. Go to **Settings** → **Customization** → **Customize Form**.
2. Select **Stock Entry** as the doctype to customize.
3. Create a **new section** to group all SRV-related fields.
4. Inside this section, add the custom fields required for SRV (such as identifiers, manufacturer details, QA information, etc.).
5. Mark the SRV number as **unique and autoname** so that each entry is traceable.
6. Set important fields like **Expiry Date** and **Screening Level** as **mandatory** to enforce quality checks.
7. Configure dropdowns (e.g., **QA Status**, **Screening Level**) with the allowed values.
8. Save and reload the form to apply changes.

add Inword number

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings** → **Workflow** → **New Workflow**.
2. Set:
 - **Workflow Name:** SRV Workflow
 - **Document Type:** Stock Entry
 - **Condition:** doc.purpose == "Material Receipt"
3. Click **Save**.

2) Add Workflow States

1. In the **States** table, add these rows:
 - **Draft**
 - **Submitted**
 - **QA Verified**
 - **Approved**
 - **Binned**

3) Add Workflow Transitions

1. In the **Transitions** table, add the following step by step:

a) Draft → Submitted

- **Action:** Submit
- **Allowed Role:** Stores Staff
- **Next State:** Submitted

b) Submitted → QA Verified

- **Action:** QA Validate
- **Allowed Role:** QA Team
- **Next State:** QA Verified

c) QA Verified → Approved

- **Action:** Approve
- **Allowed Role:** Stores Approver
- **Next State:** Approved

d) Approved → Binned

- **Action:** Bin Allocation
- **Allowed Role:** Warehouse Operator
- **Next State:** Binned

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply whenever **Stock Entry** has **Purpose = Material Receipt**.

3. Permission Rules

- **Stores Staff:** Create/Edit SRV.
- **QA Team:** Update QA status.
- **Stores Approver:** Final approval & posting.

- **Warehouse Operator:** Assign bins.

Implement via **Role Permission Manager**.

4. Binning (Warehouse Hierarchy)

- Warehouse Structure: **Warehouse** → **Room** → **Shelf** → **Bin**
- Add **Link Field** to Stock Entry Detail for Bin Assignment.
- On **Bin Allocation**, update **Stock Ledger Entry**.

5. Reports

a) SRV Register

- Report Type: List Report
- Fields: SRV Number, Supplier, Project, QA Status, Workflow State.

b) Component Expiry Report

- Query Report: Filter by expiry_date < today + 90.

c) Binning Status Report

- Show all SRVs in “Approved” state without Bin location.

d) QA Inspection Report

- Group by qa_status, show component metadata.

6. Integration Hooks

- **PO Integration:** Fetch PO details (po_number, supplier, project) via fetch_from.
- **Item Master Validation:** Match MDB PN & Manufacturer with Item doctype.
- **Radiation Module:** Add field radiation_category → mandatory at receipt.
- **Quality Module:** Screening Level + QA Approval → enforced at workflow.

8.1.1.3 SRN (Store Return Note) Module

The SRN module extends ERPNext’s Material Return Entry (Stock Entry) to handle the return of unused or screened components back into inventory. It ensures traceability to the original issue (SIV), proper binning, and Cardex updates.

1. Doctype Customization

Base Doctype

- **Stock Entry (Material Return Purpose).**
- Extend this with **custom fields** for SRN tracking.

Steps

1. Go to **Settings → Customization → Customize Form**.
2. Select **Stock Entry** as the doctype to customize.
3. Create a **new section** to group all SRN-related fields.
4. Inside this section, add the custom fields required for SRN (such as reference number, linked SIV, posting details, QA info, and Cardex references).
5. Mark the **SRN Reference Number** as **unique and autoname** so that each return entry is traceable.
6. Add a **checkbox field** for special posting cases (e.g., QA returns requiring form prints).
7. Save and reload the form to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings → Workflow → New Workflow**.
2. Set:
 - **Workflow Name:** SRN Workflow
 - **Document Type:** Stock Entry
 - **Condition:** doc.purpose == "Material Return"
3. Click **Save**.

2) Add Workflow States

- Draft
- Submitted
- QA Validated
- Approved
- Binned / Completed

3) Add Workflow Transitions

a) Draft → Submitted

- **Action:** Submit
- **Allowed Role:** Stores Executive
- **Next State:** Submitted

b) Submitted → QA Validated

- **Action:** QA Validate
- **Allowed Role:** QA Inspector
- **Next State:** QA Validated

c) QA Validated → Approved

- **Action:** Approve
- **Allowed Role:** Stores Approver
- **Next State:** Approved

d) Approved → Binned/Completed

- **Action:** Bin Allocation
- **Allowed Role:** Stores Executive
- **Next State:** Binned/Completed

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply whenever Stock Entry has **Purpose = Material Return**.
3. For **Special Posting cases**, configure workflow to require **Form Print** before closure.

3. Permission Rules

- **Stores Executive:** Create/Edit SRNs, perform binning.
- **QA Inspector:** Validate screening-related SRNs.
- **Stores Approver:** Final approval.
- **System Admin:** Handle exceptions and overrides.

Implement via **Role Permission Manager**.

4. Binning (Warehouse Hierarchy)

- Warehouse structure follows: **Warehouse** → **Room** → **Shelf** → **Bin**.
- Add a **Link field** to Stock Entry Detail for bin assignment.
- On **Bin Allocation**, update both **Stock Ledger Entry** and the **Cardex system**.
- Allow **deletion of SRN entries** if invalid or duplicate (with logging).

5. Reports

a) SRN Register

- Report Type: List Report
- Shows SRNs by Project, Linked SIV, QA status, Workflow State.

b) Pending Binning SRNs

- List all SRNs in **Approved** state without assigned bins.

c) Special Posting Report

- Report showing SRNs processed with **special posting flag** and associated **form prints**.

d) Cardex Update Log

- Shows mapping between SRN entries and Cardex updates (new vs. updated items).

6. Integration Hooks

- **SIV Module:** Each SRN references the original SIV for traceability.
- **Cardex System:** Updates or creates Cardex entries when components are returned.
- **QA / Screening Lab:** “Screened At” field links returns to QA inspection records.
- **Inventory Module:** Stock Ledger is updated once binning is complete.

8.1.1.4 SIV (Store Issue Voucher) Module

The SIV module extends ERPNext's Material Issue (Stock Entry) to manage the controlled issuance of components for projects, PCB design, production kits, on-box/system mounting, and returns. It ensures FIFO traceability, project linkage, and workflow approvals.

1. Doctype Customization

Base Doctype

- **Stock Entry (Material Issue Purpose).**
- Extend this with **custom fields** for SIV tracking.

Steps

1. Go to **Settings → Customization → Customize Form**.
2. Select **Stock Entry** as the doctype to customize.
3. Create a **new section** to group all SIV-related fields.
4. Inside this section, add the custom fields required for SIV (such as request type, program code, linked part number, card reference, pipeline flag, etc.).
5. Mark the **SIV number** as **unique and autoname** so that each issue entry is traceable.
6. Add fields for **Date Code, Expiry Date, and QA validations** to enforce compliance.
7. Configure dropdowns (e.g., **Request Type**) with predefined values like Normal, Specific, PCB, KIT, On-Box, On-System.
8. Save and reload the form to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings → Workflow → New Workflow**.
2. Set:
 - **Workflow Name:** SIV Workflow
 - **Document Type:** Stock Entry
 - **Condition:** doc.purpose == "Material Issue"
3. Click **Save**.

2) Add Workflow States

- Draft
- Submitted
- QA/Controller Validation
- Approver Decision
- Issued
- Completed / Returned

3) Add Workflow Transitions

a) Draft → Submitted

- **Action:** Submit
- **Allowed Role:** Request Initiator (Engineer/Planner/Designer)
- **Next State:** Submitted

b) Submitted → QA/Controller Validation

- **Action:** QA Validate
- **Allowed Role:** Component Controller / QA
- **Next State:** QA/Controller Validation

c) QA/Controller Validation → Approver Decision

- **Action:** Approve / Reject
- **Allowed Role:** Approver (Project Manager / Stores Authority)
- **Next State:** Approver Decision

d) Approver Decision → Issued

- **Action:** Final Issue
- **Allowed Role:** Stores Executive
- **Next State:** Issued

e) Issued → Completed / Returned

- **Action:** Close / Return
- **Allowed Role:** Stores Executive
- **Next State:** Completed or Returned

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply whenever Stock Entry has **Purpose = Material Issue**.
3. Configure **Pipeline logic** to keep items in **Staged state** for up to 15 days, with auto-return if unused.

3. Permission Rules

- **Request Initiator (Engineer/Planner/Designer):** Create Draft requests.
- **Component Controller / QA:** Validate stock, fields, and FIFO compliance.
- **Approver (Project Manager / Stores Authority):** Approve or reject issuance.
- **Stores Executive:** Perform final issue and posting to stock ledger.
- **System Admin:** Handle exceptions, overrides, and auto-returns.

Implement via **Role Permission Manager**.

4. FIFO & Binning Rules

- Enforce **FIFO logic** during issuance to select batches in correct order.
- Support **KIT Requests:** Issue grouped items while updating available stock (New Stock = Total – Kitted – PL Qty).

- Support **On-Box / On-System requests**: Capture mount or mould details before issue.
- Add **pipeline flag** to hold staged stock before final issue.
- Returns should be **linked back to original SIV** and update both **SRN and Cardex** entries.

5. Reports

a) SIV Register

- Report Type: List Report
- Shows project-wise and request-type-wise issued SIVs.

b) KIT Issuance Report

- Report summarizing all KIT requests and stock deductions.

c) Pending Approval Requests

- Report listing all SIVs awaiting approver action.

d) FIFO Compliance Report

- Report validating that stock was issued according to FIFO.

e) SIV Return Report

- Report linking returns back to original SIVs and associated SRNs.

f) Pipeline Report

- Shows staged items pending issue or auto-return.

6. Integration Hooks

- **Project-Card Configurations** → Ensure PCB requests are tied to project cards.
- **Inventory Module** → Deduct issued stock and enforce FIFO batch selection.
- **QA Module** → Validate screening and expiry before issuing.
- **SRN Module** → Manage returns linked against original SIVs.
- **BOM Integration** → Allow PCB/production requests to auto-fetch items based on BOM.

8.1.1.5 Project–Card Configuration Module

The Project–Card Configuration Module extends ERPNext’s Project and BOM doctypes to manage the hierarchical setup of projects, subsystems, packages, and cards at URSC. It ensures proper traceability, approval workflows, and integration with component issuance.

1. Doctype Customization

Base Doctypes

- **Project (ERPNext Core)** – for project registration.
- **Task (ERPNext Core)** – optional for activity tracking.
- **BOM (ERPNext Core)** – links component requirements to cards.

Custom Doctypes

- **Subsystem** (linked to Project).
- **Package** (linked to Subsystem).
- **Card** (Normal, On-Box, On-System types).

- **Card Approval Routing** (CPD, ECAD, PD/DPD).

Steps

1. Go to **Settings** → **Customization** → **Customize Form**.
2. Select **Project** as the doctype and add fields such as Project Reference Number, PD Staff Number, Program Code, etc.
3. Create **custom doctypes** for Subsystem, Package, and Card, each linked hierarchically.
4. Add fields for **card type**, **approval status**, and **constraint validation**.
5. Create a **routing doctype** for CPD, ECAD, and PD/DPD approvals.
6. Save and reload the forms to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings** → **Workflow** → **New Workflow**.
2. Set:
 - **Workflow Name:** Project–Card Workflow
 - **Document Type:** Card (custom doctype)
 - **Condition:** doc.card_type in ["Normal","On Box","On System"]
3. Click **Save**.

2) Add Workflow States

- Draft
- Submitted
- Under Review
- Approved
- Released / On Hold / Withdrawn

3) Add Workflow Transitions

a) Draft → Submitted

- **Action:** Submit
- **Allowed Role:** Design Engineer / Planner
- **Next State:** Submitted

b) Submitted → Under Review

- **Action:** Send for Review
- **Allowed Role:** CPD / ECAD Coordinators
- **Next State:** Under Review

c) Under Review → Approved

- **Action:** Approve
- **Allowed Role:** PD/DPD Authorities
- **Next State:** Approved

d) Approved → Released

- **Action:** Release
- **Allowed Role:** Project Administrator
- **Next State:** Released

e) Approved → On Hold

- **Action:** Hold
- **Allowed Role:** QA / Production Controller
- **Next State:** On Hold

f) Approved → Withdrawn

- **Action:** Withdraw
- **Allowed Role:** QA / Production Controller
- **Next State:** Withdrawn

g) **Withdrawn → Replacement Added** (for replacement cards)

- **Action:** Add Replacement
- **Allowed Role:** Design Engineer
- **Next State:** Approved

4) **Save and Publish**

1. Click **Save**.
2. Workflow now enforces approvals for card configuration.

3. Permission Rules

- **Project Administrator / Program Manager:** Create Projects, Subsystems, and Packages.
- **Design Engineer / Planner:** Add Cards and submit for approvals.
- **QA / Production Controller:** Hold, Release, or Withdraw cards.
- **CPD / ECAD Coordinators:** Technical validation and approval.
- **PD/DPD Authorities:** Final management approval.

Implement via **Role Permission Manager**.

4. Card & Package Configuration

- Create **Subsystems** under Projects.
- Add **Packages** within Subsystems.
- Add **Cards** (Normal, On-Box, On-System) within Packages.
- Enable **Hold/Release/Withdraw** logic on cards.
- For **Replacement Cards**, allow addition only after the original card is withdrawn.
- Support **SAT model**: Create packages as New, From Existing, or Production Package.

5. Reports

a) **Project Hierarchy Report**

- Project → Subsystem → Package → Card.

b) **Card Status Report**

- Shows Draft, Approved, On Hold, Withdrawn, Replacement cards.

c) **Approval Tracking Report**

- Tracks approvals across CPD, ECAD, and PD/DPD.

d) **Configuration History Report**

- Shows card replacements, withdrawals, and approvals.

6. Integration Hooks

- **Inventory Module:** Link cards to issued components (via SIV).
- **BOM Integration:** Cards tied directly to BOM entries for manufacturing traceability.
- **QA Module:** QA team manages Hold/Release based on compliance checks.
- **Approvals Module:** Integrates ECAD, CPD, and PD/DPD workflows into the card lifecycle.

8.1.1.6 Master Database Management Module

8.1.1.6.1 Overview

The Master Database Management (MDB) Module in ERPNext manages all core master data entities required for engineering, production, inventory, procurement, and system administration. Masters are the foundational reference data tables that drive transactional processes and ensure data consistency across the system.

This module ensures controlled creation, validation, and updates of part numbers, quality levels, vendors, package styles, users, and roles. It provides traceability, approval workflows, audit logging, and alerts for all master data changes.

8.1.1.6.2 Master pages in the Project

The following master pages are implemented in ERPNext for this project:

- **Part Master (Component Master)** – Central record of all parts/components. Stores links to BPN, Quality Level, Package Style, Vendor, Link Part Number, and other attributes such as Footprint and Base Part Number.
- **Basic Part Number (BPN) Master** – Maintains standardized basic part numbers with version control and traceability across revisions.
- **Quality Level Master** – Defines QA compliance levels (e.g., QL-1, QL-2) for use in approval workflows and part release validation.
- **Package Style (PStyle) Master** – Catalog of approved package styles to enforce BOM consistency and ensure only validated package formats are used in design and inventory.
- **Link Part Number Master** – Maintains cross-referenced part numbers for alternate or equivalent components, supporting supplier and engineering traceability.
- **Vendor / Manufacturer Master** – Contains approved vendors and manufacturers, along with their linkage to parts and approval history.
- **User Master** – ERPNext standard, extended with access controls and role assignments for managing user accounts securely.
- **Role Master** – ERPNext standard, extended with workflow approvals and audit logging to ensure controlled system access and segregation of duties.

Each master page supports **controlled entry, validation, approval workflows, and field-level rules** to ensure consistency and compliance across ERPNext.

8.1.1.6.2.1 Part Master (Component Master)

The **Part Master (Component Master)** is the central repository of all parts/components used in engineering and production. It ensures traceability by linking each part to its associated basic part number, quality level, vendor, package style, and link part number.

Functions:

1. Define and register new parts with unique identifiers.
2. Link parts to BPN, Quality Level, Package Style, Vendor, and Link Part Number records.

3. Control revisions and approvals before parts are released into production.

Key Fields:

- Part Code / Component Code (unique)
- Part Name / Component Name
- Basic Part Number (BPN) reference
- Quality Level reference
- Package Style reference
- Vendor/Manufacturer reference
- Link Part Number reference
- Footprint and Base Part Number

Usage:

- Used in BOMs, Inventory, SRNs, and SIVs.
- Prevents duplication of parts and enforces compliance before release.

8.1.1.6.2.2 Basic Part Number (BPN) Master

The **BPN Master** maintains a standardized list of Basic Part Numbers with revision tracking.

Functions:

1. Enforce standardized numbering conventions.
2. Manage versioning and revision control.
3. Provide BPN references for Part Master entries.

Key Fields:

- BPN Code (unique)
- Description
- Version Number / Effective Date
- Linked Part Masters

Usage:

- Provides consistency in part identification.
- Supports traceability across BOMs and revisions.

8.1.1.6.2.3 Quality Level Master

The **Quality Level Master** defines compliance levels (e.g., QL-1, QL-2) that must be associated with parts.

Functions:

1. Define QA compliance codes.
2. Enforce quality requirements during part creation.
3. Provide linkage for approvals and audits.

Key Fields:

- Quality Level Code
- Description / Compliance Requirement
- Approval Status

Usage:

- Enforces QA validation before part release.
- Supports audit and compliance reporting.

8.1.1.6.2.4 Package Style (PStyle) Master

The **Package Style Master** defines the approved set of package styles available in the system.

Functions:

1. Maintain standardized package style data.
2. Provide linkage for Part Master records.
3. Restrict BOMs to use only approved package styles.

Key Fields:

- Package Style Code (unique)
- Package Style Name / Description
- Associated Part Masters

Usage:

- Enforces consistency in packaging definitions.
- Reduces BOM errors by controlling package style assignments.

8.1.1.6.2.5 Link Part Number Master

The **Link Part Number Master** maintains cross-references to related part numbers for traceability.

Functions:

1. Store approved link part numbers.
2. Enable association of multiple identifiers for the same component.

3. Provide traceability across suppliers and systems.

Key Fields:

- Link Part Number Code (unique)
- Associated Part Master(s)
- Vendor / Manufacturer Reference

Usage:

- Supports alternate part number references in procurement and design.
- Helps in cross-referencing equivalent or interchangeable parts.

8.1.1.6.2.6 Vendor / Manufacturer Master

The **Vendor Master** manages the approved list of vendors and manufacturers.

Functions:

1. Register new vendors/manufacturers.
2. Link approved vendors to specific parts.
3. Maintain vendor approval history.

Key Fields:

- Vendor Code / Name
- Contact Details
- Approved Parts Linked
- Status (Active/Inactive)

Usage:

- Ensures procurement only from authorized suppliers.
- Provides supplier performance data for audits.

8.1.1.6.2.7 User Master

The **User Master** defines ERPNext users and their associated roles.

Functions:

1. Create and manage ERPNext user accounts.
2. Assign access levels and permissions.

3. Track user activity via audit trails.

Key Fields:

- User ID / Profile Information
- Assigned Roles
- Access Rights / Restrictions

Usage:

- Provides secure and role-based ERPNext access.
- Ensures user accountability in workflows.

8.1.1.6.2.8 Role Master

The **Role Master** defines system roles, workflows, and permissions.

Functions:

1. Create and manage role definitions.
2. Configure workflow transitions tied to roles.
3. Maintain audit logs for approvals.

Key Fields:

- Role Name
- Permissions and Doctype Access
- Workflow States and Transitions
- Audit History

Usage:

- Enforces segregation of duties.
- Provides structured workflow approvals for masters and transactions.

8.1.1.6.3 Process of Creating and Managing Masters

The creation and maintenance of masters follows a consistent process:

1. **Navigation** – User goes to the required Master Page (e.g., Part Master).
2. **Record Creation** – Click **New** to add a new master entry.
3. **Data Entry** – Provide all required fields (e.g., Component Code, Category, Package Style).
4. **Validation** – System automatically checks for:

- Duplicate entries (unique fields).
 - Missing mandatory attributes.
 - Invalid references to other masters.
5. **Workflow Routing** – Masters are routed through approval workflows depending on type (see section 8.1.1.6.4).
 6. **Approval** – Approvers validate correctness before release.
 7. **Commit** – Once approved, the master becomes active and usable across ERPNext.

1. Doctype Customization

Base Doctypes

- **User** (ERPNext Standard, extended with access controls).
- **Role** (ERPNext Standard, extended for workflow approvals).

Custom Doctypes

- **Part Master** – Central table for all parts with links to BPN, Quality, Package, Vendor.
- **Basic Part No (BPN)** – Maintains master list of basic part numbers with version control.
- **Quality Level** – Defines QA compliance levels (e.g., QL-1, QL-2).
- **Package Style** – Approved package styles for BOM consistency.
- **Vendor / Manufacturer** – Manages approved vendors and supplier-part linkages.

Steps

1. Go to **Settings** → **Customization** → **Customize Form**.
2. Create new doctypes for **Part Master**, **Basic Part No**, **Quality Level**, **Package Style**, **Vendor**.
3. Add link fields in **Part Master** to connect to BPN, Quality, Package, and Vendor.
4. Extend the **User doctype** with fields for linked role and menu/submenu access rights.
5. Extend the **Role doctype** with audit and workflow fields for role approvals.
6. Save and reload to apply changes.

2. Workflow Setup

1) Create a New Workflow (Part No Update)

1. Go to **Settings** → **Workflow** → **New Workflow**.
2. Set:
 - **Workflow Name:** Part No Update Workflow
 - **Document Type:** Part Master
 - **Condition:** doc.status == "Draft"
3. Click **Save**.

2) Add Workflow States

- Draft
- QA Review
- Approved
- Released

3) Add Workflow Transitions

a) Draft → QA Review

- **Action:** Send for QA Review
- **Allowed Role:** Designer
- **Next State:** QA Review

b) QA Review → Approved

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- **Action:** Approve
- **Allowed Role:** QA Officer
- **Next State:** Approved
- c) **Approved → Released**
 - **Action:** Release
 - **Allowed Role:** Project Manager
 - **Next State:** Released
- 4) **Create a New Workflow (User Role Management)**
 1. Go to **Settings → Workflow → New Workflow**.
 2. Set:
 - **Workflow Name:** User Role Workflow
 - **Document Type:** Role
 - **Condition:** doc.status == "Draft"
 3. Click **Save**.
- 5) **Add Workflow States**
 - Draft
 - Security Review
 - Approved
- 6) **Add Workflow Transitions**
 - a) **Draft → Security Review**
 - **Action:** Send for Security Review
 - **Allowed Role:** System Admin
 - **Next State:** Security Review
 - b) **Security Review → Approved**
 - **Action:** Approve Role
 - **Allowed Role:** Security Reviewer
 - **Next State:** Approved

3. Permission Rules

- **Part Master Admin:** Create/Update Part Master and BPNs.
- **QA Approver:** Approve Part Nos, Quality Levels, and Package Styles.
- **Vendor Manager:** Manage Vendor/Manufacturer approvals.
- **System Admin:** Create roles, manage access, and review security.
- **Project Manager:** Final release approvals.

Implement via **Role Permission Manager**.

4. Alerts & Notifications

- Trigger alerts when:
 - A **BPN** is updated or released.
 - A **role** is modified or newly approved.
- Notifications go to **QA, Project Managers, Vendors, and System Admin**.

5. Reports

- a) **Part Master Register**
 - Shows Part Nos, BPNs, Quality Levels, Package Styles, Vendors.
- b) **BPN Version Report**

- Tracks version history and updates of Basic Part Numbers.
- c) **Quality Level Report**
 - Lists QL codes with descriptions and references.
- d) **Vendor Approval Report**
 - Shows approved vendors and their associated parts.
- e) **User Role Audit Report**
 - Tracks creation, changes, and approvals of roles.

6. Integration Hooks

- **Project–Card BOM** → Only approved Part Nos, Package Styles, and Vendors can be linked.
- **Inventory & Procurement** → Vendor mappings and part master data are used in purchasing.
- **Quality Module** → Quality Level validation enforced before release.
- **User & Role Management** → Integrated with ERPNext’s permission framework.

8.1.1.7 Project–Card Component List (BOM) Module

The Project–Card Component List module extends ERPNext’s **Bill of Materials (BOM)** doctype to manage the structured component list for each **Project–Card**. It ensures traceability, controlled review workflows, and prevents modifications once fabrication begins.

1. Doctype Customization

Base Dotypes

- **Bill of Materials (BOM)** – parent doctype for assemblies.
- **BOM Item (Child Table)** – sub-components under BOM.

Customizations

- On **BOM (Parent)**: Add fields for Project (Link) and Card (Link).
- On **BOM Items (Child Table)**: Add fields for Position and Attributes (Key-Value / JSON).

Steps

1. Go to **Settings** → **Customization** → **Customize Form**.
2. Select **BOM** as the doctype to customize.
3. Create a **new section** to group all Project–Card fields.
4. Add **Project (Link → Project)** and **Card (Link → Card)** fields.
5. In **BOM Item child table**, add **Position** and **Attributes** fields for detailed mapping.
6. Save and reload the forms to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings** → **Workflow** → **New Workflow**.
2. Set:
 - **Workflow Name**: Project–Card BOM Workflow

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- **Document Type:** BOM
 - **Condition:** doc.is_active == 1
- 3. Click **Save**.
- 2) Add Workflow States**
 - Draft
 - Review
 - Approved
 - Fabricated
- 3) Add Workflow Transitions**
 - a) **Draft → Review**
 - **Action:** Submit for Review
 - **Allowed Role:** Designer
 - **Next State:** Review
 - b) **Review → Approved**
 - **Action:** Approve
 - **Allowed Role:** Reviewer
 - **Next State:** Approved
 - c) **Approved → Fabricated**
 - **Action:** Mark as Fabricated
 - **Allowed Role:** Project Manager
 - **Next State:** Fabricated
- 4) Save and Publish**
 - 1. Click **Save**.
 - 2. Workflow will now apply whenever a **Project–Card BOM** is created.
 - 3. Once status = **Fabricated**, editing is **disabled**.

3. Permission Rules

- **Designer:** Create/Edit BOMs in Draft only.
- **Reviewer:** Validate and approve/reject BOMs.
- **Project Manager:** Final approval before fabrication.
- **Fabrication Team:** View-only access (no edits).

Implement via **Role Permission Manager**.

4. Restrictions

- **Fabrication Lock:** Once BOM is in Fabricated state, no edits are allowed.
- **Duplication:** Approved BOMs can be duplicated for new Project–Cards.
- **Bulk Editing:** Enable controlled updates via **Data Import Tool** or custom Bulk Edit form.

5. Reports

- a) **Project–Card BOM Register**
 - Report Type: List Report
 - Shows BOMs linked with Projects and Cards.
- b) **Pending Reviews Report**
 - List of BOMs awaiting Reviewer action.
- c) **Fabricated BOM Report**

- Shows BOMs locked after fabrication.

d) **Bulk Update Log**

- Audit trail of mass component updates.

6. Integration Hooks

- **Item Master:** Validate that all BOM components exist in Item Master.
- **Project Module:** Link BOMs with projects for traceability.
- **Card Doctype:** Associate BOMs with individual cards.
- **Workflow Module:** Enforce approvals and fabrication restrictions.
- **Inventory/Stock:** Approved BOMs are referenced in material planning and stock allocation.

8.1.1.8 Radiation Module

The **Radiation Module** extends ERPNext's **Quality** and **Project** modules to manage radiation exposure data for **EEE components** at the **Card / Package** level. It ensures traceability of radiation test history, compliance with program-specific requirements, and controlled approval workflows.

1. Doctype Customization

Base DocTypes

- **Project / Card** – to represent the hierarchy (Project → Subsystem → Package → Card).
- **Item** – component master linked to Card/Package.

Custom Doctype

- **Radiation Record** – captures radiation test details.

Steps

1. Go to **Settings** → **Customization** → **Customize Form**.
2. Create a **new doctype Radiation Record**.
3. Add fields for Component/Card Reference, Package No., Test Type, Radiation Level, Test Date, Test Facility, QA Status, and Approved By.
4. Enable file attachment for uploading lab reports, certificates, and images.
5. Save and reload to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings** → **Workflow** → **New Workflow**.

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2. Set:
 - **Workflow Name:** Radiation Record Workflow
 - **Document Type:** Radiation Record
 - **Condition:** doc.status == "Draft"

3. Click **Save**.

2) Add Workflow States

- Draft
- Submitted
- QA Reviewed
- Approved

3) Add Workflow Transitions

a) Draft → Submitted

- **Action:** Submit
- **Allowed Role:** Designer
- **Next State:** Submitted

b) Submitted → QA Reviewed

- **Action:** QA Review
- **Allowed Role:** QA Team
- **Next State:** QA Reviewed

c) QA Reviewed → Approved

- **Action:** Approve
- **Allowed Role:** Radiation Approver
- **Next State:** Approved

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply whenever a **Radiation Record** is created.

3. Permission Rules

- **Designer:** Create Radiation Records.
- **QA Team:** Review and update QA Status.
- **Radiation Approver:** Final approval and sign-off.

Implement via **Role Permission Manager**.

4. Record Management

- **Radiation Data Entry:** Users record radiation test parameters against Cards/Packages.
- **Lab Report Upload:** Attach scanned lab reports, images, or certificates.
- **Traceability:** Each Radiation Record links back to Project → Subsystem → Package → Card.
- **Failure Linkage:** If a failure is observed, trigger integration with **FA/DPA modules**.

5. Reports

a) Radiation Record Register

- Report Type: List Report
- Shows all entries by Project, Card, Package, Test Date.

b) Radiation Compliance Report

- Tracks Cards/Packages with approved radiation data.

c) Pending Radiation Approvals

- Shows records awaiting QA or Approver validation.

d) KPI Dashboard

- Displays exposure statistics by program, subsystem, or date range.

6. Integration Hooks

- **Project–Card Configurations:** Link records to Project hierarchy.
- **Item Master (MDB):** Validate components and radiation categories.
- **Failure Analysis (FA) Module:** Auto-create FA requests from failed radiation records.
- **DPA Module:** Link radiation-tested parts for further analysis.
- **Quality Module:** Enforce QA review before approval.

8.1.1.9 Failure Analysis (FA) Module

The **Failure Analysis (FA) Module** manages the lifecycle of component failure investigations – from request creation, analysis, result documentation, and corrective actions – while ensuring traceability and compliance.

1. Doctype Customization

Base Doctype

- **Failure Analysis Request (Custom Doctype)** – core FA tracking.

Steps

1. Go to **Settings** → **Customization** → **Doctype** → **New Doctype**.
2. Create a **new doctype: Failure Analysis Request**.
3. Add fields for Component ID, Failure Description, Requested By, Root Cause Analysis, Corrective Action, Status, and Linked Documents.
4. Enable file attachments for storing reports, images, and test results.
5. Save and reload the form to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings** → **Workflow** → **New Workflow**.
2. Set:
 - **Workflow Name:** FA Workflow
 - **Document Type:** Failure Analysis Request
 - **Condition:** doc.status == "Draft"

3. Click **Save**.

2) Add Workflow States

- Draft
- In Analysis
- Completed / Closed

3) Add Workflow Transitions

a) Draft → In Analysis

- **Action:** Assign for Analysis
- **Allowed Role:** Designer / QA
- **Next State:** In Analysis

b) In Analysis → Completed

- **Action:** Submit Analysis
- **Allowed Role:** FA Engineer
- **Next State:** Completed

c) Completed → Closed

- **Action:** Approve Closure
- **Allowed Role:** Project Manager
- **Next State:** Closed

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply whenever an FA Request is created.

3. Permission Rules

- **Designer / QA:** Create FA requests.
- **FA Engineer:** Update analysis details, RCA, and corrective actions.
- **Project Manager:** Approve and close FA requests.

Implement via **Role Permission Manager**.

4. Failure Analysis Process

- **FA Request Creation:** User raises request with component details.
- **Assignment:** FA Engineer assigned for investigation.
- **Analysis:** Document failure type, root cause, and corrective action. Attach supporting reports and images.
- **Corrective Action:** Record RCA and corrective steps taken.
- **Closure:** Project Manager reviews and closes the FA case.

5. Reports

a) FA Register

- Report Type: List Report
- Displays all FA requests with status and linked components.

b) Component Failure Report

- Shows failure trends and counts per component/project.

c) Corrective Action Effectiveness Report

- Tracks whether corrective actions reduced recurring failures.

6. Integration Hooks

- **Item Master / BOM:** Link FA requests to failed components.
- **Project Module:** Maintain traceability of failures to specific projects.
- **DPA Module:** Trigger DPA requests if destructive analysis is required.
- **Document Storage:** Store reports, RCA documents, and supporting images.

8.1.1.10 Destructive Physical Analysis (DPA) Module

The **Destructive Physical Analysis (DPA) Module** manages destructive testing of components. It ensures proper request handling, execution, result documentation, and traceability with related FA cases.

1. Doctype Customization

Base Doctype

- **DPA Request (Custom Doctype)** – core doctype for tracking DPA requests.

Steps

1. Go to **Settings** → **Customization** → **Doctype** → **New Doctype**.
2. Create a **new doctype: DPA Request**.
3. Add a section to capture DPA-related details.
4. Inside this section, configure fields such as Component ID, Test Performed, Laboratory Reference, Results & Conclusion, Linked FA Request, and Status.
5. Enable **file attachments** for lab reports, test results, and images.
6. Mark the **Status** field with controlled values (Draft, In Testing, Results Available, Closed).
7. Save and reload the form to apply changes.

2. Workflow Setup

1) Create a New Workflow

1. Go to **Settings → Workflow → New Workflow**.
2. Set:
 - **Workflow Name:** DPA Workflow
 - **Document Type:** DPA Request
 - **Condition:** doc.status == "Draft"
3. Click **Save**.

2) Add Workflow States

- Draft
- In Testing
- Results Available
- Closed

3) Add Workflow Transitions

a) Draft → In Testing

- **Action:** Assign for Testing
- **Allowed Role:** QA / Designer
- **Next State:** In Testing

b) In Testing → Results Available

- **Action:** Upload Results
- **Allowed Role:** Lab Engineer
- **Next State:** Results Available

c) Results Available → Closed

- **Action:** Approve Closure
- **Allowed Role:** Project Manager
- **Next State:** Closed

4) Save and Publish

1. Click **Save**.
2. The workflow will now apply whenever a DPA Request is created.

3. Permission Rules

- **QA / Designer:** Create and submit DPA requests.
- **Lab Engineer:** Upload results, reports, and conclusions.
- **Project Manager:** Approve and close requests.

Implement via **Role Permission Manager**.

4. DPA Process

- **DPA Request Creation:** QA/Designer raises request with component details.
- **Testing Assignment:** Request is assigned to lab/test team.
- **Testing Execution:** Lab performs destructive tests and uploads results.
- **Results Validation:** QA team reviews test outcomes.
- **Closure:** Project Manager finalizes and closes the request.

5. Reports

a) DPA Register

- Report Type: List Report
- Shows all DPA requests with component, project, and status.
- b) **Lab Results Report**
 - Summarizes conclusions of destructive tests.
- c) **Linked FA Report**
 - Combined view of FA and related DPA cases for traceability.

6. Integration Hooks

- **Failure Analysis Module:** Link DPA cases with FA requests.
- **Item Master / BOM:** Ensure failed component traceability.
- **Project Module:** Track DPA at the project level.
- **Document Storage:** Store lab reports, test images, and conclusions.

8.1.1.11 Views and Reports for All Modules

The reporting framework provides analytical, operational, and MIS views across all modules, ensuring transparency, traceability, and audit readiness.

1. Report & Dashboard Setup

Base Tools

- **Query Reports** – For SQL-based tabular reports.
- **Script Reports** – For customized logic and data processing.
- **Dashboards (Standard + Custom)** – For graphical and consolidated views.

Steps

1. Go to **Settings** → **Customization** → **Reports**.
2. Create **Query Reports** for structured data like SRV, SRN, and SIV summaries.
3. Create **Script Reports** for advanced logic (e.g., FIFO compliance, radiation exposure).
4. Build **Dashboards** under *Settings* → *Dashboards* for consolidated KPIs.
5. Add **filters** for critical attributes (Project, Card, Link Part Number, Expiry Date, Package Style, Quality Level).
6. Enable **export options** for Excel and PDF.
7. Configure **drill-down navigation** from summary → transaction level.
8. (Optional) Enable **Scheduled Report Delivery** → set frequency and recipients.

2. Permission Rules

SDD

- **Project Managers:** Access to project-level and BOM vs. Issued reports.
- **QA Team:** Access to Quality, Radiation, FA, and DPA reports.
- **Stores Staff / Approvers:** Access to stock movement and binning status.
- **System Admin:** Access to audit logs and activity dashboards.

3. Integration Hooks

- **Inventory Module** – Pulls stock data from SRV, SRN, SIV.
- **Project-Card Module** – Links BOMs and project configurations for planning.
- **Quality Modules (Radiation, FA, DPA)** – Fetches test results and compliance data.
- **User & Role Management** – Ensures access to reports based on roles.

9. System Design

9.1 System Architecture

- **High-level architecture:**
ERPNext Core + Custom Modules + Integrations
- **Components:**
 - ❖ **ERPNext standard modules:** Buying, Selling, Stock, Assets, Manufacturing, Quality, Users, ERPNext Settings, ERPNext Integrations, Projects, Home.
 - ❖ **Custom Modules / Doctypes:** SRV (Store Receipt Voucher), SRN (Store Return Note), SIV (Store Issue Voucher), HDD Project, Stock Verification, Radiation Module, Screening Lab.
 - ❖ **Integrations:** Email notifications only (no accounting / GL).
- **Diagrams:**
 - ❖ **Context Diagram:** Users → ERPNext System → External email system / notifications.
 - ❖ **Component Diagram:** ERPNext core + custom modules + database + SLE (Stock Ledger Entry) + store/warehouse logic.

9.2 Module Descriptions

This is where you should put the explanations of all **existing ERPNext modules** (Accounting, Stock, Sales, HR, etc.), in the short descriptive format you showed.

9.2.1 Home

1. Purpose: Central dashboard with role-based access, shortcuts, and KPIs.

2. Navigation: Automatically loads after login.

3. Features:

- Role-based workspaces (Stores, QA, Engineers, Project Managers)
- Quick links: SRV Entry, Pending QA, Binning Queue, Work Orders, Project Board
- Recently viewed docs; assigned tasks

4. Flow:

1. User logs in → Workspace opens.
2. Click shortcut → start process (e.g., SRV Entry).

5. Data Model: Workspace, Number Cards, Dashboard Charts

6. Validations: Role-based access

7. Roles & Notifications: N/A

8. Reports/KPIs: Open assignments, pending approvals, binning backlog

9.2.2 Buying

1. Purpose:

The Buying module manages the procurement cycle of materials, components, and services from vendors. It ensures that all purchases follow a controlled process: creating Purchase Orders (POs), optionally recording Supplier Quotations, and completing goods receipt into Stores through **SRV Entry**.

2. Navigation

ERPNext Path:

Home → Buying

3. Features

- **Supplier Master:** Maintain supplier details (contacts, addresses, tax IDs, approval status).
- **Supplier Quotation (Optional):** Record quotations received from suppliers, if needed for price comparison.

- **Purchase Order (PO):** Formal order to a supplier with items, quantities, and delivery dates.
- **PO Status Tracking:** Ordered, Partially Received, Completed, or Cancelled.
- **Integration with SRV:** On delivery, PO lines link directly to **SRV Entry** for controlled receipt and QA checks.

4. Flow

- **Identify Requirement** → Trigger from Projects, Manufacturing, or manual request.
- **Create RFQ** → Send to selected suppliers.
- **Record Supplier Quotation** → Capture supplier offers, lead times, and prices.
- **Compare & Select** → Choose supplier using comparison tool.
- **Issue PO** → Submit Purchase Order to chosen supplier.
- **Track PO** → Monitor status (pending delivery, partial receipt, completed).
- **Delivery Arrives** → **Stores creates Purchase Receipt:**
 - ❖ Enter received qty, QA checks
 - ❖ **Data Model**
- **Supplier:** Holds supplier details and approval level.
- **RFQ:** Header (supplier list, date), Child Table (items required).
- **Supplier Quotation:** Header (supplier, validity), Child Table (items, rates, terms).
- **Purchase Order:** Header (supplier, terms, delivery date), Child Table (items, qty, rate).
- **Address/Contact:** Linked to Supplier for communications.
- **Linkages:**
 - ❖ **RFQ ↔ Supplier Quotation**
 - ❖ **Supplier Quotation ↔ Purchase Order**
 - ❖ **Purchase Order ↔ SRV Entry (line-level mapping)**

5. Validations

- **PO must be Submitted** before creating an SRV Entry.
- **Items in SRV Entry must exist in PO** (no mismatched items).
- **Qty checks:**
 - ❖ Received Qty $\leq$ Ordered Qty (unless allowed with approval).
 - ❖ Partial receipts tracked properly (balance qty remains open).
- **Supplier checks:** Only approved suppliers allowed if approval workflow is enabled.
- **Expiry/QA rules:** If item requires QA verification, SRV Entry cannot be posted without QA result.

6. Roles & Notifications

- **Roles:**
 - ❖ **Buying User:** Can create RFQs, Supplier Quotations, and draft POs.
 - ❖ **Buying Manager:** Approves POs, supplier selection.
 - ❖ **Stores User:** Creates SRV Entry against PO.
 - ❖ **QA User:** Performs inspection, accepts/rejects items.
- **Notifications:**
 - ❖ Email Buyer when QA rejects or shortage occurs during SRV Entry.
 - ❖ Reminder email for overdue POs not yet delivered.

7. Reports & KPIs

- **Supplier On-Time Delivery %** = $(\text{On-time deliveries} \div \text{Total deliveries}) \times 100$
- **PO Cycle Time** = Date of PO submission – Date of RFQ creation
- **Pending PO Lines** = Number of open PO lines not fully received (tracked via SRV Entry)
- **Supplier Performance Report:** Combine price, quality (QA rejection %), and delivery timeliness.

9.2.3 Stock / Inventory

1. Purpose

The Stock module ensures **complete control of materials** in your system.
For EEE Component Management, it:

- Tracks every **component, assembly, and consumable**.
- Controls where items are kept (**Stores** like Main Store, QA Hold, Quarantine).
- Records all **movements**: from supplier delivery, QA acceptance, binning, to project issues.
- Maintains **traceability** with batch numbers, serial numbers, and expiry/relifing dates.
- Provides **visibility** through Stock Ledger and Balance reports

2. Navigation

- **ERPNext Path:**

Home → Stock

- **Shortcuts available:**
 - ❖ **Item** (EEE component master)
 - ❖ **Stock Entry** (Issue, Receipt, Transfer)
 - ❖ **Warehouse (Store)** setup
 - ❖ **Stock Ledger / Balance**
 - ❖ **Stock Reconciliation**

3. Features

1. Item Management

- Define item codes, names, categories (IC, capacitor, resistor, connector, etc.).
- Set batch/serial tracking rules.
- Configure expiry and relifing dates.

- Link to Quality templates (QA, Screening, Radiation).

2. Stores (Warehouses)

- Logical separation of stock:
 - Inbound Dock → temporary holding.
 - QA Hold → inspection pending.
 - Main Store → accepted and usable.
 - Quarantine → rejected or blocked items.
- In your setup: **Warehouses = Stores**.

3. Stock Entry

- **Material Issue** → send components to projects/departments.
- **Material Transfer** → move between Stores.
- **Material Receipt** → rarely used, since SRV covers receipts.

4. Batch & Serial Tracking

- **Batch**: Bulk items with one batch number (resistors, capacitors).
- **Serial**: Unique tracking for high-value ICs or modules.

5. Stock Reconciliation

- Matches physical stock with system balances.
- Used in **Stock Verification module** to correct discrepancies.

6. Reports

- **Stock Ledger**: Line-by-line history (every SRV, issue, transfer).
- **Stock Balance**: Current stock at each Store.
- **Stock Ageing**: Detects long-stored/obsolete items.

4. Flow

1. Receipts

- Materials delivered by supplier.
- **SRV Entry → QA inspection → SRV Post (accepted qty only) → SRV Binn (assign bin inside Main Store).**
- Rejected items go to *Quarantine* or are returned to supplier.

2. Issues

- Stores issue items to **Projects** using **Stock Entry (Material Issue)**.
- Issues reference Project/Task for traceability.
- Serial/batch info is carried to keep history clear.

3. Transfers

- Components moved between Stores (e.g., Main Store → Lab Store).
- Recorded via **Stock Entry (Material Transfer)**.

4. Reconciliation

- During **Stock Verification**, physical counts are matched with ERPNext balances.
- Variances corrected through **Stock Reconciliation**.

5. Data Model

Core Doctypes:

- **Item:** All EEE components (resistor, IC, capacitor, etc.).
- **Warehouse (Store):** Physical/logical storage (Main Store, QA Hold, Quarantine).
- **Bin (ERPNext):** Auto-summary of item qty in each Store.
- **Batch:** Tracks batch-controlled items.
- **Serial No:** Tracks individually tagged items.
- **Stock Entry:** Transaction doc for issues, transfers, receipts.

- **Stock Ledger Entry (SLE):** Immutable log of every movement.

Custom Doctypes:

- **Storage Bin:** Sub-location within Store (rack, shelf, bin code).
- **SRV Binn:** Assigns items to specific bins.

6. Validations

- **Batch/Serial enforcement** → Cannot post without required details.
- **Negative stock prevention** → System blocks issues if qty < 0 (unless allowed).
- **Company/Store consistency** → Each movement tied to one company, valid Stores.
- **Expiry/Relifing enforcement** → Blocks issuing expired or invalid items.

7. Roles & Notifications

- **Stores User:** Create Stock Entries, SRV Binn, run reports.
- **Stores Manager:** Approve reconciliations, manage transfers.
- **QA User:** View stock balances for inspection records.
- **Notifications:**
 - Low stock alerts (when below reorder level).
 - Alerts for items nearing expiry.
 - Reminders for unbinned stock after SRV.

8. Reports & KPIs

1. Stock Accuracy %

- Formula:
$$\frac{1 - (|\text{System} - \text{Physical}| \div \text{System}) \times 100}{100}$$

- Measures accuracy between ERPNext and physical counts.

2. Stock Turnover Ratio

- Formula:
$$\frac{\text{Cost of Goods Issued}}{\text{Average Inventory Value}}$$
- Measures how quickly inventory is consumed.

3. Stock Ageing Report

- Shows slow-moving or obsolete items.

4. Value of Discrepancies

- Total adjustment value after stock verification.

5. Bin Utilization % (Custom)

- Formula:
$$\frac{\text{Occupied Bins}}{\text{Total Bins}} \times 100$$
- Ensures optimal usage of storage space.

In ERPNext, every sale, purchase, or stock movement updates the **General Ledger**.

Navigation: Home > Accounting

Main Features:

- Chart of Accounts
- Journal Entries
- Accounts Payable/Receivable
- Financial Reports (Balance Sheet, P&L, Trial Balance)

9.2.4 Quality

1. Purpose

The Quality module ensures that **all items meet defined inspection and compliance standards** before being accepted into stock or dispatched.

For EEE component management, it plays a critical role by:

- Verifying incoming items at receipt (via SRV).

- Checking in-process and finished goods during manufacturing.
- Recording non-conformances and rejections for supplier and production feedback.
- Ensuring only **QA-approved items** move forward in Stores and Projects.

2. Navigation:

Home → Quality

3. Features

1. Quality Inspection Templates

- Predefined checklists and parameters (e.g., dimension check, resistance test, radiation level, relifing check).
- Standardize QA process across items.

2. Incoming Inspections

- Triggered when items arrive (via **SRV Entry** or Purchase Receipt).
- Ensures only approved items are posted to stock.

3. In-Process Inspections

- Performed during **Manufacturing operations** (linked to Job Cards).
- Helps detect issues early before assembly is complete.

4. Outgoing Inspections

- Before dispatch or project delivery.
- Ensures items leaving Stores meet compliance.

5. Non-Conformance Reporting (NCR)

- Record defects or failures.
- Assign corrective actions (scrap, return to supplier, rework).
- Escalate repeat supplier issues for evaluation.

6. Integration with Other Modules

- **Buying/SRV** → Incoming QA at receipt.
- **Stock** → Block rejected items in Quarantine Store.
- **Manufacturing** → In-process checks at operations.

4. Flow

1. **Create Inspection Template** → Define standards for item type (e.g., IC testing, capacitor voltage rating).
2. **Perform Inspection** → At SRV (incoming), during production (in-process), or at delivery (outgoing).
3. **Record Results** → Enter pass/fail data, measurements, observations.
4. **Accept or Reject:**
 - Accepted qty moves into Stores.
 - Rejected qty moved to *Quarantine* or returned.
5. **NCR Logging** → If rejected, document issue and corrective action.

5. Data Model

- **Quality Inspection:** Parent doc recording results.
- **Quality Inspection Template:** Master list of inspection parameters.
- **Non-Conformance Report (NCR):** Log of rejected items/issues.
- **Links:**
 - Quality Inspection ↔ Item
 - Quality Inspection ↔ SRV Entry / Job Card / Delivery Note

6. Validations

- Items requiring QA **cannot be posted to stock** until inspection is complete.
- Measurements must be filled where mandatory.
- Batch/serial must be linked if required.

- NCR required for all rejected quantities.

7. Roles & Notifications

- **QA User:** Perform inspections, enter results.
- **QA Manager:** Approve/reject inspections, close NCRs.
- **Stores User:** Cannot bin items until QA approves.
- **Notifications:**
 - ❖ Email Stores on QA approval/rejection.
 - ❖ Notify Buyer when supplier items are rejected.
 - ❖ Alerts for pending inspections.

8. Reports & KPIs

- **First Pass Yield (FPY)** = $(\text{Accepted qty} \div \text{Inspected qty}) \times 100$
- **QA Rejection Rate** = $(\text{Rejected qty} \div \text{Inspected qty}) \times 100$
- **NCR Frequency by Supplier** = Count of NCRs per supplier over time
- **Inspection Lead Time** = Average time taken to complete inspections
- **Trend Analysis:** Repeated failures by item type or supplier

9.2.4 Selling

The **Selling module** handles the complete sales cycle, from managing leads and opportunities to quotations, orders, and delivery.

Navigation: Home > Selling

Main Features:

- Customer records.
- Sales Orders and Quotations.

SDD

- Delivery Notes.
- Sales Invoices.

9.2.5 Manufacturing

The Stock module manages warehouses and inventory movements.

In ERPNext, stock is tracked across multiple warehouses, with FIFO and valuation.

Navigation: Home > Stock

Main Features:

- Material Receipt (SRV), Issue (SIV), Return (SRN)
- Stock Transfers between warehouses
- Batch and Serial tracking
- Stock Ledger and Balance reports

9.2.6 Users

1. Purpose

The Users and Permissions module provides **access control, approvals, and work assignment** in ERPNext.

For your EEE Component & Inventory Management project, it ensures:

- Only authorized staff can perform tasks like creating SRV Entries, approving QA inspections, or reconciling stock.
- Roles define responsibilities (Stores, QA, Projects, Admin).
- Permissions restrict access by Company, Store, or Document type.
- Assignments allow tasks (SRV review, inspection, approvals) to be routed to the right person.

2. Navigation:

Home → Users and Permissions

3. Features

- **User:** Represents a system account (with login, email, role assignments).
- **Role:** Defines access rights (e.g., Stores User, QA User, Buyer, Project Manager).
- **Role Profile:** Predefined role bundles for quick user setup.

- **User Permission:** Record that restricts access to specific Companies, Stores, or Projects.
- **Assignment Rule:** Automatically assigns documents (like SRV, Inspection, or Tasks) to a user or group based on conditions.

4. Flow

1. Create User

- Enter name, email, department, company.

2. Assign Roles

- Example: Stores User, QA Manager, Project Engineer.

3. Apply User Permissions

- Restrict by Company, Store, or Project so users see only relevant data.

4. Assignments

- Documents such as **SRV Entries, Quality Inspections, or Tasks** can be assigned to specific users for action.

5. Approvals & Actions

- Users perform assigned tasks (approve SRV, complete inspections, close tasks).

5. Data Model

- **User:** Core doctype with login and profile details.
- **Role:** Master list of roles.
- **Role Profile:** Grouping of roles.
- **User Permission:** 6Links User ↔ Restricted entity (Company, Store, Project).
- **Assignment:** Tracks which documents are assigned to which users.

6. Validations

- **Least-Privilege Principle:** Users only get access to what they need.

- **Company/Store Rules:** Must match assigned Company and Store for transactions.
- **Approval Dependencies:** Certain actions (e.g., SRV Posting) may require Manager role.

7. Roles & Notifications

- **Admin:** Manages users, roles, permissions.
- **Stores Manager/User:** Access to Stock Entries and SRV.
- **QA Manager/User:** Access to Quality Inspections and NCRs.
- **Project Manager/User:** Access to project-related tasks.
- **Notifications:**
 - Automatic email when a document (SRV, Inspection, Task) is assigned.
 - Reminder notifications for pending approvals or overdue assignments.

8. Reports & KPIs

- **Open Assignments Report:** Shows documents currently assigned and pending action.
- **Approval SLA Tracking (optional):** Measures time taken for approvals (e.g., average SRV approval time).
- **User Access Audit Report:** Who has access to what, ensuring compliance.

9.2.7 ERPNext Settings

1. Purpose

The Settings module provides the **administration and configuration backbone** of ERPNext. For your EEE Component & Inventory Management project, this module ensures:

- Company details, fiscal year, and currency are correctly defined.
- Templates for printing (e.g., SRV, Inspection Reports) and emails are standardized.
- Approval workflows (for SRV Posting, QA inspections, Project approvals) are enforced.
- System-wide defaults (time zone, naming series, notification settings) are centrally controlled.

2. Navigation:

Home → Settings

3. Features

1. Company

- Define company details (name, address, domain).
- Link company to Stores/Warehouses and Projects.

2. Global Defaults

- Base currency (e.g., INR).
- Timezone (e.g., Asia/Kolkata).
- Default naming series for documents (e.g., SRV-0001).

3. Print Settings

- Configure global print styles.
- Customize layouts for SRV, Quality Inspection, NCR reports.

4. Workflow

- Create approval flows for documents:
 - SRV → Requires Stores Manager approval.
 - QA → Requires QA Manager approval.
- Define workflow states, transitions, and approvers.

5. Email Account

- Configure SMTP/IMAP for official company email.
- Used for system-generated emails (notifications, approvals, alerts).

6. Notification

- Define triggers for auto-emails (e.g., SRV rejection, inspection pending).

- Use templates for consistency.

4. Flow

1. **Company Setup** → Create company master record.
2. **Set Defaults** → Configure currency, timezone, naming series.
3. **Configure Email** → Set up SMTP/IMAP accounts for outgoing/incoming mail.
4. **Create Workflows** → Build approval rules for SRV, QA, and NCR.
5. **Define Notifications** → Set automated messages for key events (rejections, pending approvals).
6. **Test & Validate** → Verify email delivery and workflow transitions.

5. Data Model

- **Company:** Master record for organization details.
- **Global Defaults:** Currency, timezone, and defaults.
- **Email Account:** Stores SMTP/IMAP credentials.
- **Notification:** Template + trigger + recipient.
- **Workflow:** Document type, states, transitions, approvers.

6. Validations

- **Test Email Account** before activation to ensure SMTP/IMAP works.
- **Workflow checks:** Each state must have valid transitions and approvers.
- **Company defaults** must be set before transactions can be posted.

7. Roles & Notifications

- **System Manager:** Responsible for all configuration.
- **Notifications:**
 - Emails as configured for workflow approvals.

- Alerts for pending tasks (e.g., SRV awaiting approval, QA rejection)

8. Reports & KPIs

- No standard KPIs here (settings is administrative).
- Optional monitoring:
 - Email error logs.
 - Workflow SLA reports (time taken for approvals).

9.2.8 ERPNext Integrations

1. Purpose

This module connects ERPNext with email servers (SMTP/IMAP) to enable **system notifications and communication**.

For your EEE Component & Inventory Management project, its purpose is to:

- Automatically send alerts (e.g., SRV created, QA rejected, binning completed, Work Order ready).
- Ensure users are kept informed without logging in to ERPNext constantly.
- Maintain delivery logs for auditing.

2. Navigation:

Home → Integrations → Email Account

3.Features

1. Email Account Setup

- Configure outgoing SMTP accounts (e.g., noreply@neemus.com).
- Configure incoming IMAP accounts (optional, for inbound mail handling).
- Support multiple accounts for different roles or modules.

2. Email Templates

- Standardized templates for SRV creation, QA inspection results, NCR rejections, binning updates, WO readiness.
- Consistent subject lines and branding.

3. Notifications

- Triggered automatically when specific events occur:
 - SRV Entry created.
 - QA Inspection approved/rejected.
 - Binning completed.
 - Work Order ready for production.

4. Delivery Logs

- ERPNext keeps records of each notification (success/failure).
- Helps troubleshoot email delivery issues.

4. Flow

1. **Configure Email Account** → Set SMTP/IMAP details, test connection.
2. **Create Email Templates** → Define subjects and body (with placeholders like {item_name}, {qty}).
3. **Define Notifications** → Link templates to events (e.g., SRV creation).
4. **Trigger Event** → User action (like submitting SRV Entry) generates email.
5. **Notification Sent** → Email sent to role-specific users.
6. **Check Logs** → Admin monitors delivery success/failure.

5. Data Model

- **Email Account:** SMTP/IMAP configuration.
- **Notification:** Rules linking events → recipients → email templates.
- **Email Template:** Standard format for messages.
- **Email Log:** Record of outgoing/incoming emails, delivery status.

6. Validations

- **Connectivity:** SMTP/IMAP tested before saving account.

- **Authentication:** Secure login (SSL/TLS) required.
- **Spam-safety:** Ensure subject/body doesn't trigger spam filters.
- **Template Linking:** Every notification must reference a valid template.

7. Roles & Notifications

- **Administrator:** Configures accounts, templates, notifications.
- **System Users:** Receive emails based on their role and document assignments.
- **Examples:**
 - Buyer notified when SRV rejected.
 - QA Manager notified when NCR created.
 - Stores User notified when Work Order materials are ready.

8. Reports & KPIs

- **Notification Delivery Success %** = $(\text{Delivered} \div \text{Sent}) \times 100$
- **Email Failure Logs** = Count of failed deliveries (with reasons).
- **Pending Notifications** (if queues are used).

9.3 Custom Modules

9.3.1 SRV Module (Store Receipt Voucher)

1. Overview

SRV digitizes the receipt process against a PO with QA gating and controlled binning. Only **approved quantities** are posted and binned.

Doctypes

1. **SRV Entry** (Parent + Item Child Table)
2. **SRV Post** (Parent + Item Child Table)
3. **SRV Binn** (Bin Items asper Posted quantity)

2. Navigation

- Home → SRV → **SRV Entry**
- Home → SRV → **SRV POST**
- Home → SRV → **SRV Bin**(for pending binning queue)

3. SRV Entry — Design

1. Purpose: Capture delivery details **against the Purchase Order.**

2. Validations:

- PO must be Submitted; item must exist in PO lines.
- 11 digit INWARD number Mandatory
- Received Qty ≥ 0 ; **Accepted + Rejected = Received.**
- If Batch/Serial required → enforce capture.
- Expiry/Relifing must be valid (not expired, meets rules).
- If QA not completed → SRV Post blocked.

3. Emails:

- On *Under QA*: Email to QA.
- On *QA Rejected*: Email to Buyer & Stores.
- On *Ready to Post*: Email to Stores (posting team).

4. SRV Post — Design

1. Purpose: Post **only Accepted Qty** to stock.

How it works:

- For each SRV Entry Item with *Accepted Qty* > 0 :
 - Create a **Stock Entry (Receipt)** from *Inbound Dock* to **QA Hold** (optional) or directly to **Main Store** if QA already accepted.
 - Log success/fail per line in SRV Post.

Key Fields:

- Link to SRV Entry
- Posting Date/Time, Posting User
- Lines: Item, Accepted Qty, From Store, To Store, Status, Error Message

2. Validations:

- Process only items with *QA Accepted*.
- Do not exceed *POSTED Qty* (idempotent reruns skip already Received qty).

3. Emails:

- On completion → Notify Stores & QA: *Posting completed* or *Posting errors*.

5. SRV Binn — Design

1. Purpose: Bin posted quantities into **specific Storage Bins** inside the **Main Store**.

How it works:

- Pull a queue of **unbinned accepted quantities** (per item/batch/serial) from SRV Post results.
- User selects **Storage Bin** (custom master: Store, Aisle, Rack, Shelf, Bin Code, Capacity/Unit).
- Updates quantity left to bin; closes when fully binned.

Key Fields:

- Link to SRV Entry & SRV Post Line
- Component, Quantity to Bin, **Storage Bin** (required), Remaining Qty
- Timestamp, User

2. Validations:

- Do not exceed Binning QTY , as per SRV ENTRY and SRV POST.

3. Emails:

- On completion: notify Stores Supervisor (optional).

4. Edge Cases & Rules

- **Partial Acceptance:** Post only accepted; rejected moves to *Quarantine* or Return to Supplier (RTS) via Stock Entry.
- **Short/Excess:** Flag on SRV Entry; email Buyer; optionally create **PO Amendment Request**.
- **Multiple Batches/Serials:** Allowed at line split-level; maintain traceability.
- **Expiry/Relifing:** Block posting if invalid; allow conditional accept with manager override (logged).
- **Idempotency:** SRV Post tracks remaining to post; re-running does not duplicate.
- **Permissions:** Only QA can set accepted/rejected; only Stores can Post/Binn.

9.3.2 ADD Project Module

1. Purpose

The Project module is used to **plan, execute, and track engineering projects**. In your EEE component management system, it is customized to handle

Project → Subsystem → Package → Card hierarchy.

- Each **Card** is the smallest design or work unit.
- Cards can be linked to **Bills of Materials (BOMs)** for component requirements.
- Approval workflows ensure that only reviewed and authorized project structures move forward.

This ensures **traceability** of every component from

project planning → BOM definition → procurement → QA → stock usage.

2. Navigation:

Home → ADD Projects Module

3. Features

1. Project Hierarchy

- Define top-level Project.

- Break down into **Subsystems** → **Packages** → **Cards**.
- Each level maintains parent-child linkage.

2. Card Configuration

- Smallest unit of work.
- Linked to a **BOM** for material planning.
- Supports revision/version tracking.

3. BOM Linkage

- Cards can reference existing BOMs for assemblies/components.
- Ensures procurement and manufacturing know exactly what's required.

4. Approval Workflow

- Cards require **ECAD/CPD or Engineering Manager approval** before being used in downstream processes.
- Workflow ensures that unapproved designs cannot trigger procurement.

5. Integration with Other Modules

- **Buying:** Approved BOMs → Purchase Orders.
- **Stock:** Materials issued to Cards via Stock Entry (Material Issue).
- **QA:** Verification of components linked to Cards.

4. Flow

1. **Create Project** → Define top-level project.
2. **Add Subsystems & Packages** → Break down structure logically.
3. **Create Cards** → Define work packages, link BOMs.
4. **Approval** → Submit Cards for approval (ECAD/CPD).
5. **Procurement & Stock** → Approved Cards drive procurement, SRV receipts, and stock issues.
6. **Execution & Tracking** → Cards progress tracked (Not Started → In Progress → Completed).

5. Data Model

- **Project:** Parent document.
- **Subsystem:** Child linked to Project.
- **Package:** Child linked to Subsystem.
- **Card:** Smallest unit, linked to BOM and workflow.
- **BOM:** Linked at Card level for material requirements.
- **Approval Workflow:** Custom workflow state machine for Card approvals.

6. Validations

- BOM must be approved before linking to Card.
- Cards cannot trigger procurement unless **Approved**.
- Hierarchy integrity: Package must belong to Subsystem, Subsystem must belong to Project.
- Role-based checks: Only Engineering Managers can approve Cards.

7. Roles & Notifications

- **Project Manager:** Creates Projects, assigns responsibilities.
- **Engineer:** Creates Cards, links BOMs.
- **ECAD/CPD/Engineering Manager:** Approves Cards.
- **Stores User:** Issues components against approved Cards.
- **Notifications:**
 - Email approver when Card is submitted for approval.
 - Email Project Manager when Cards are approved or rejected.
 - Reminder for pending approvals.

8. Reports & KPIs

- **Project Progress** = (Completed Cards ÷ Total Cards) × 100
- **Approval Lead Time** = Average time taken to approve Cards.
- **Open vs Approved Cards**: Monitors design bottlenecks.
- **Project Material Readiness**: % of BOM items available in stock vs pending procurement.

9.3.3 Radiation Module

The Radiation module ensures **compliance for radiation-sensitive EEE components**. It provides a structured process to:

- Record component radiation measurements.
- Upload supporting lab test reports.
- Enforce **threshold rules** that block unsafe items from being posted to stock (SRV Post) or delivered (Delivery Note).
- Track trends and turnaround time (TAT) of lab inspections.

This guarantees that only **safe and compliant components** move into Stores and Projects.

2. Navigation:

Home → Quality → Radiation Module

3. Features

1. Measurement Recording

- Enter component ID, batch/serial, and measured radiation levels.
- Record date, lab technician, and equipment used.

2. Lab Report Upload

- Attach PDF/Excel reports from external radiation testing labs.
- Linked directly to the item and inspection record.

3. Threshold Rules

- Define safe radiation limits per item category.
- Automatically block SRV Post or Delivery Note if exceeded.
- Provide override option for QA Manager with justification.

4. Review & Approval

- QA User reviews lab data and uploaded reports.
- QA Manager approves or rejects the batch.
- Rejected items move to *Quarantine Store*.³

4. Flow

1. **Record Measurement** → Create radiation record against item/batch/serial.
2. **Upload Lab Report** → Attach supporting test file.
3. **Review** → QA User checks recorded values vs threshold.
4. **Approve/Reject:**
 - **Approve** → Item allowed for SRV Post / Delivery Note.
 - **Reject** → Item blocked, NCR logged if required.
5. **Compliance Enforcement** → System ensures only approved items can be posted/issued.

5. Data Model

- **Radiation Record:** Parent doc with item, batch, measurement, lab details.
- **Lab Report:** File attachment linked to Radiation Record.
- **Threshold Rules:** Configurable per item or item group.
- **Linkages:**
 - Radiation Record ↔ Item / Batch / Serial No.
 - Radiation Record ↔ SRV Entry / Delivery Note.

6. Validations

- Radiation Record required before SRV Post for radiation-sensitive items.
- Measurements must be within threshold (else block).
- Uploaded report mandatory if rule requires.
- QA Manager approval required for any override.

7. Roles & Notifications

- **Lab Technician:** Creates Radiation Record, uploads lab report.
- **QA User:** Reviews data and performs initial check.
- **QA Manager:** Approves/blocks items, manages overrides.
- **Notifications:**
 - Email QA Manager when new record is submitted.
 - Email Buyer if items are rejected for radiation compliance.

8. Reports & KPIs

- **Items Over Threshold:** List of all batches/serials exceeding safe limits.
- **Trend Analysis:** Radiation level trends per supplier or component category.
- **Lab Turnaround Time (TAT):** Average time between receipt of sample and report approval.

9.3.4 Screening Lab

1. Purpose

The Screening Lab module manages **screening tests for EEE components** to ensure reliability and compliance before stock posting or project use.

It allows:

- Submitting screening requests for components.
- Assigning tests to specific labs.
- Recording and approving results.
- Linking test outcomes directly to **SRV Entries** and **Items**.

- Triggering **SRN (Store Return Note)** for items that fail screening, ensuring they are returned or quarantined.

2. Navigation:

Home → Quality → Screening Lab

3. Features

1. Screening Requests

- Create request for one or more items received (linked to SRV Entry).
- Define test type (burn-in, thermal, vibration, electrical, etc.).
- Attach supplier/lab documents.

2. Lab Assignment

- Assign request to an internal or external lab.
- Track assignment status (Assigned → In Progress → Completed).

3. Result Recording

- Enter test measurements, Pass/Fail outcome.
- Upload lab test reports (PDF/Excel).
- Auto-link results to the Item and Batch/Serial.

4. QA Approval

- QA User reviews test results.
- QA Manager approves or rejects.
- Rejected items are routed to **SRN module** for return/disposal.

5. Integration with SRN (Store Return Note)

- Failed items automatically trigger SRN draft.
- SRN tracks rejected quantity and ensures items are physically returned to supplier or quarantined.

4. Flow

1. **Request** → Stores/QA raises Screening Request for item(s) in SRV Entry.
2. **Assign** → Screening Lab Manager assigns request to lab.
3. **Test** → Lab performs screening tests.
4. **Result** → Lab uploads report and marks Pass/Fail.
5. **QA Approval** →
 - If **Pass** → Items move to Main Store (via SRV Post & Binning).
 - If **Fail** → System triggers **SRN** to remove failed qty from stock and route for return/disposal.

5. Data Model

- **Screening Request:** Parent doc with item details, linked SRV Entry.
- **Screening Lab Assignment:** Tracks which lab is responsible.
- **Screening Result:** Measurements, pass/fail decision, uploaded files.
- **SRN (Store Return Note):** Document for managing return/disposal of failed items.
- **Linkages:**
 - Screening Request ↔ SRV Entry / Item / Batch.
 - Screening Result ↔ QA Approval.
 - Failed Result ↔ SRN.

6. Validations

- Screening Request cannot close without QA Manager approval.
- Failed results must generate SRN entry before request is marked “Closed”.
- Batch/Serial numbers must match SRV Entry for traceability.
- Test report upload mandatory for compliance.

7. Roles & Notifications

- **Stores User:** Creates Screening Request.
- **Lab Technician:** Records results, uploads reports.
- **QA User:** Reviews data.
- **QA Manager:** Approves or rejects items.
- **Notifications:**
 - Email Lab Technician when assigned.
 - Email QA Manager when results are submitted.
 - Email Buyer if SRN is created due to failure.

8. Reports & KPIs

- **Turnaround Time (TAT)** = Result Date – Request Date.
- **Pass Rate %** = (Passed Items ÷ Total Tested) × 100.
- **Fail Rate %** = (Failed Items ÷ Total Tested) × 100.
- **Rework %** = (Reworked ÷ Total Tested) × 100.
- **Supplier Performance Report** = Linked to failure rates from supplier deliveries.